Data Preprocessing Project: Sentiment Analysis

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1 Introduction

In the world of online education, Massive Open Online Courses (MOOCs) have become super popular. They offer a variety of courses and allow people from all over the world to learn at their own pace. One key way we understand how well these courses are working is through the feedback from learners, like reviews and comments. But here's the hitch: going through all these comments manually is a real tough job. This is where the idea of sentiment analysis for MOOC learner reviews comes in.

2 Define the Problem

2.1 Problem and context Statement

The central challenge is the labor-intensive manual review of Massive Open Online Courses (MOOCs) learner feedback, intensified by the growing volume of reviews. Streamlining through sentiment analysis is essential. Navigating through a substantial pile of comments is akin to searching for treasures in darkness. Advanced Natural Language Processing (NLP) techniques during data preprocessing, like tokenization and lowercasing, are crucial. The question is: How can we use advanced NLP and data preprocessing to effectively automate sentiment analysis and derive actionable insights for continual MOOC improvement?

3 Data Collection

The dataset for this project encompasses a vast collection of over 9 million comments sourced from diverse Massive Open Online Courses (MOOCs). These comments represent the perspectives of individuals engaged in various courses, offering a broad spectrum of experiences and languages. The dataset is a valuable resource for analyzing learner sentiments, providing insights into course content, experiences, and preferences.

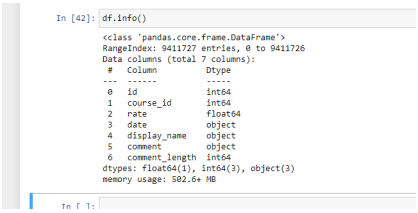
Key features in the dataset include:
ID: Unique identifier for each comment.
Course ID: Identifier for the specific MOOC course.
Rate: Rating given by the learner.
Date: Timestamp indicating when the comment was made.
Display Name: Identifier for the user who posted the comment.
Comment: The actual text-based feedback provided by the learner.

4 Exploratory Data Analysis (EDA)

4.1 Basic Statistics

We can start performing statistical analysis using various pandas functions. Here are some common statistical operations we can perform:

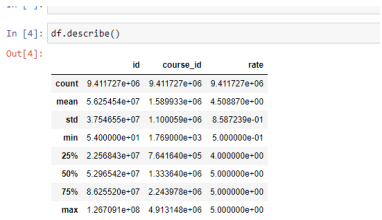
• **Descriptive Statistics:** We use the describe() method to get a summary of basic statistical measures for each numerical column in our dataset.



```
In [42]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9411727 entries, 0 to 9411726
Data columns (total 7 columns):
 #   Column          Dtype
---  ------          -----
 0   id              int64
 1   course_id       int64
 2   rate            float64
 3   date            object
 4   display_name    object
 5   comment         object
 6   comment_length  int64
dtypes: float64(1), int64(3), object(3)
memory usage: 502.6+ MB
```

Figure 1: Informations on Data Frame



```
In [4]: df.describe()
Out[4]:
```

	id	course_id	rate
count	9.411727e+06	9.411727e+06	9.411727e+06
mean	5.625454e+07	1.589933e+06	4.508870e+00
std	3.754655e+07	1.100059e+06	8.587239e-01
min	5.400000e+01	1.769000e+03	5.000000e-01
25%	2.256843e+07	7.641640e+05	4.000000e+00
50%	5.296542e+07	1.333640e+06	5.000000e+00
75%	8.625520e+07	2.243978e+06	5.000000e+00
max	1.267091e+08	4.913148e+06	5.000000e+00

Figure 2: Describe Data Frame

• **Counting and Frequency:**
Counting and analyzing unique values doesn't directly help detect outliers, but it's essential to consider the possibility of outliers in our dataset. Techniques such as statistical measures and dedicated algorithms are commonly employed for outlier detection, while visualizations and imputation methods are effective for handling missing values. Unique value analysis, focusing on the frequency of occurrences, contributes to understanding data distribution but isn't the primary tool for identifying outliers or missing values, and filter for some columns when it's needed.

• **Checking Missing Values:**
Missing values in textual data, such as comments or feedback, can impact the quality of NLP analyses, potentially leading to biased or incomplete results. Addressing missing values beforehand allows for appropriate handling or deletion strategies, ensuring a more accurate and reliable NLP process.

• **Correlation:**
Basically correlation calculation visualization helps identify relationships between text-related features (e.g., sentiment, text length) and other numerical feautures (e.g., ratings, date), guiding NLP analysis and preprocessing decisions.

• **Unique Values:**
Checking unique values is essential for categorical data. Specifically, for IDs, detecting and removing duplicates, especially in course IDs, maintains data

integrity and prevents biases in analysis.

4.2 Visualization

Visualization refers to the creation and interpretation of graphical representations of data. Visualization is a crucial component of EDA because it allows analysts to gain insights into the patterns, trends, and relationships within a dataset more effectively than through raw data alone. Visualizations provide a way to explore and communicate complex information in a more accessible and intuitive manner.

• **Visualize the Distribution of Rate:** A Distribution of Course Ratings is a bar chart will be used to visualize the distribution of course ratings, providing insights into the prevalence of each rating. This will help understand the overall sentiment of the reviews.

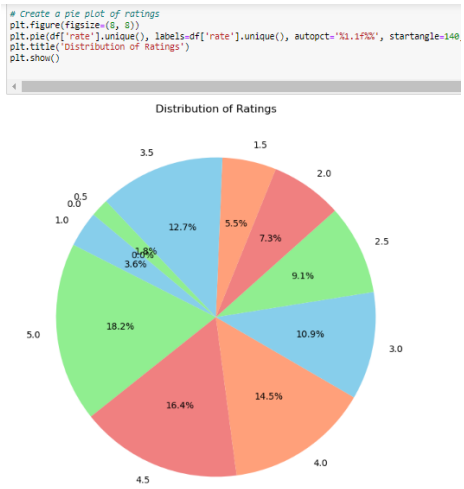


Figure 3: Distribution of rates

• **Visualization of the most Common words in The comments :**

Analyzing the most commonly used words in comments based on their sentiment (negative, neutral, or positive) offers valuable insights. This visualization helps us understand distinct language patterns for each sentiment category, guiding improvements in courses with lower ratings and informing content creators about the nature of reviews. This focused analysis emphasizes the importance of tailored text preprocessing in revealing meaningful language patterns in our data.



Figure 4: Visualization of the most used words in positive comments



Figure 5: Visualization of the most used words in negative comments

5 Preprocessing :

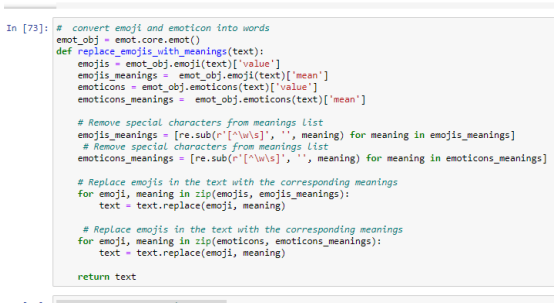
5.1 Data Cleaning :

During the data preprocessing phase, several steps were undertaken to ensure the quality and relevance of the dataset. First, null comments were removed to eliminate any instances lacking valuable feedback. Duplicate entries with identical IDs were also addressed, streamlining the dataset to avoid redundancy. Additionally, the "ID" and "Display Name" columns were excluded from the analysis, as they were deemed unnecessary for the NLP tasks at hand.

5.1.1 NLP Text Cleaning:

To prepare the text data for Natural Language Processing (NLP), we conducted a comprehensive cleaning process. This involved eliminating special characters, spaces, and blanks to enhance comment readability. Removal of special characters and numbers aimed to minimize noise and irrelevant information in the dataset. Standardizing the text by converting it all to lowercase was crucial for maintaining consistency and mitigating case sensitivity issues during analysis. Additionally, we implemented a function to eliminate links using regular expressions and a dedicated pattern to identify and remove URLs. To refine the text corpus further, we employed a function to strip away HTML tags. Specific textual treatments, such as transforming "don't" to "do not," were applied, ensuring a uniform and more meaningful representation of the language.

Also in our NLP augmentation phase, we extended our text preparation beyond cleaning by some essential other NLP techniques. Tokenization, the process of breaking down text into smaller units such as words or phrases, enhances our ability to analyze and extract features. Lemmatization, which reduces words to their base or root form, ensures consistency in meaning across different forms of the same word. Additionally, the removal of stop words, common words with limited contribution to meaning, helps reduce noise and improves the efficiency of our analysis. These techniques collectively aim to



```
In [73]: #  convert emoji and emoticon into words
emot_obj = emot.core.emot()
def replace_emojis_with_meanings(text):
    emojis = emot_obj.emoji(text)['value']
    emojis_meanings =  emot_obj.emoji(text)['mean']
    emoticons = emot_obj.emoticons(text)['value']
    emoticons_meanings =  emot_obj.emoticons(text)['mean']

    # Remove special characters from meanings list
    emojis_meanings = [re.sub(r'[^\w\s]', '', meaning) for meaning in emojis_meanings]
     # Remove special characters from meanings list
    emoticons_meanings = [re.sub(r'[^\w\s]', '', meaning) for meaning in emoticons_meanings]

    # Replace emojis in the text with the corresponding meanings
    for emoji, meaning in zip(emojis, emojis_meanings):
        text = text.replace(emoji, meaning)

     # Replace emojis in the text with the corresponding meanings
    for emoji, meaning in zip(emoticons, emoticons_meanings):
        text = text.replace(emoji, meaning)

    return text
```

Figure 6: Converting emojis to Words

refine the text data, facilitating more effective and insightful NLP analysis in our dataset.

Lastly, we converted emojis into words. This action, enhancing the clarity and informativeness of the text, aligns with our feature engineering efforts to enrich the data for better analysis and model development.

5.2 Feature Engineering :

In our feature engineering phase, we made key enhancements to our dataset to optimize it for building models. Initially, we identified and excluded comments not in English, ensuring language consistency for effective analysis.

Next, we organized the data into distinct groups based on various rating conditions. This categorization, coupled with the addition of a sentiment feature, empowers us in creating models to predict specific outcomes by considering the emotional tone of comments.

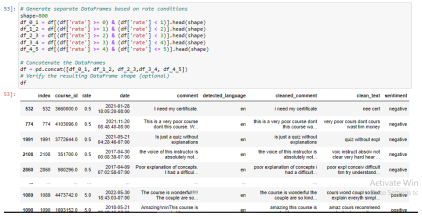


Figure 7: Sentiment Feature Based on Rating



Figure 8: Detecting Language

6 Modeling :

Modeling involves creating and training a mathematical model to make predictions or classifications based on input data. for example we have Word embedding, on the other hand, is a technique used to represent words as vectors in a continuous vector space (by giving random numbers to each word) . This method captures semantic relationships between words, enabling the model to understand contextual meanings. Word embedding is applied during the modeling phase to enhance the representation of words in the dataset, allowing the model to better understand the context and relationships between different words.

For our data we used "RoBERTa" is a language model that excels at understanding and generating contextualized representations of text, making it valuable for tasks like sentiment analysis in comments. It provides contextual embeddings, deep semantic understanding, and effective feature extraction.



Figure 9: Word Embadding

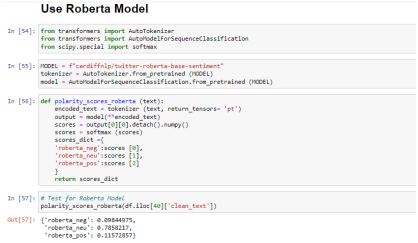


Figure 10: Roberta Model

In the training phase following modeling, we input our preprocessed data into the selected machine learning or deep learning model. Here, we specify the model architecture and predictions, The overarching goal is to furnish our model with the necessary knowledge to make precise predictions when presented with new and unseen data.

- Our Training Accuracy : 0.8995 - Our Testing Accuracy : 0.8965

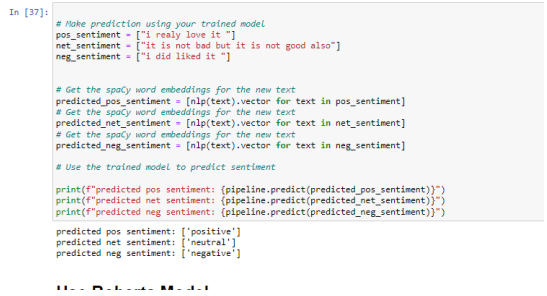


Figure 11: Predicted Model